

Insurance Cost Prediction

A detailed machine learning project report

From data understanding to a reusable prediction pipeline

This report documents the work completed in the supplied Jupyter notebook: data inspection, cleaning, exploratory analysis, categorical encoding, model training, evaluation, model selection, feature importance, and saved-pipeline testing.

DATA USED	MODELS COMPARED	BEST TEST R ²
1,337 cleaned records	3 regression algorithms	0.9012

Key outcome

The Random Forest Regressor with max_depth=4 achieved the strongest recorded test performance. Smoking status was the most influential feature in its fitted model.

Project type: Supervised learning - regression

Target: Medical insurance charges

Source: Insurance_Cost_Prediction.ipynb

Report prepared: 10 October 2026

Evidence note: Results are transcribed from the notebook's saved outputs. Original notebook plots are reproduced. Supplementary counts are calculated from the adjacent insurance.csv after duplicate removal. The notebook has not been rerun for this report.

Executive summary & contents

The project aimed to estimate medical insurance charges from age, sex, body mass index (BMI), number of children, smoking status, and region. Because charges are numeric, the task was treated as a regression problem.

The dataset originally contained 1,338 rows and seven columns. No missing values were found. One duplicate record was removed, leaving 1,337 observations. Categorical predictors were encoded, and the cleaned data were divided into 1,069 training records and 268 test records using `random_state=42`.

Linear Regression, Decision Tree Regression, and Random Forest Regression were trained on the same split. Random Forest achieved $R^2 = 0.9012$, $MAE = 2,478.18$, and $RMSE = 4,260.66$. It reduced MAE by approximately 40.67% and RMSE by 28.47% relative to Linear Regression.

The selected approach was rebuilt as a scikit-learn pipeline combining automatic categorical encoding and Random Forest prediction. The notebook saved the pipeline to `insurance_pipeline.pkl`, reloaded it, and produced a sample prediction of 6,579.06 in dataset charge units.

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Problem, objectives & scope

Problem statement

Medical insurance charges vary across individuals. This project explores whether a small set of demographic and lifestyle variables can explain some of this variation and support a numerical estimate for a new customer profile.

Project aim

Build and evaluate a machine learning model that predicts the charges column from six customer attributes. The notebook also investigates which recorded inputs have the largest influence on the selected model.

Objectives completed

- Understand dataset dimensions, data types, descriptive statistics, and data quality.
- Remove duplicate records and inspect distributions and feature-target relationships.
- Convert categorical inputs into numerical features for regression algorithms.
- Train and compare Linear Regression, Decision Tree, and Random Forest models.
- Evaluate held-out predictions with MAE, MSE, RMSE, and R^2 .
- Identify the best model among the three evaluated approaches and inspect its feature importance.
- Package preprocessing and prediction into one saved pipeline, reload it, and test a customer example.

Inputs and output

INPUTS	OUTPUT
age, sex, bmi, children, smoker, region	Estimated value of charges

Intended use and project boundaries

The completed work is a data science prototype for estimating the recorded charge target and explaining patterns in this dataset. The notebook does not document live insurer integration, production deployment, a formal pricing policy, or a prediction interval.

Target interpretation

The target is named charges. The notebook does not establish its time period, whether it represents a premium or another billed amount, or its verified currency. This report therefore describes the output as estimated medical insurance charges in dataset units.

Dataset & descriptive statistics

The notebook loads `insurance.csv` with `pandas.read_csv()`. Initial inspection shows 1,338 observations and seven variables: six predictors and one numeric target. The dictionary below describes the fields as used in the project.

VARIABLE	TYPE	DESCRIPTION / VALUES
age	Integer	Age; observed range 18-64.
sex	Category	Recorded categories: female, male.
bmi	Float	Body mass index; observed range 15.96-53.13.
children	Integer	Number of children in the record; range 0-5.
smoker	Category	Smoking status: no, yes.
region	Category	northeast, northwest, southeast, southwest.
charges	Float	Continuous prediction target; range 1,121.87-63,770.43.

Numerical summary saved in the notebook

STATISTIC	AGE	BMI	CHILDREN	CHARGES
Mean	39.21	30.66	1.09	13,270.42
Std. deviation	14.05	6.10	1.21	12,110.01
Minimum	18	15.96	0	1,121.87
25th percentile	27	26.30	0	4,740.29
Median	39	30.40	1	9,382.03
75th percentile	51	34.69	2	16,639.91
Maximum	64	53.13	5	63,770.43

These statistics come from `dataset.describe()` before the duplicate was removed, so they describe the original 1,338 rows. The EDA, training, and later counts use 1,337 cleaned records.

What the summary suggests

Mean charges exceed the median, which is consistent with the long right tail in the histogram. The large standard deviation indicates substantial variation in charges. This makes error measures and plots important alongside a single model score.

Data quality & project workflow

CHECK	NOTEBOOK FINDING	ACTION
Dimensions	1,338 rows x 7 columns	Inspect <code>dataset.head()</code> , <code>info()</code> , <code>describe()</code> .
Missing values	0 in every column	No imputation was needed.
Duplicate records	1 exact duplicate	Remove using <code>drop_duplicates()</code> .
After removal	1,337 rows; 0 duplicates	Use the cleaned dataset for EDA and modeling.
Duplicate column names	0	No column-name correction needed.

Cleaning decisions

The notebook retains the observed high-charge records and the full BMI range. It does not perform outlier deletion, scaling, log transformation of charges, or additional feature construction. Removing the duplicate avoids counting the same observed record twice.

Workflow followed

1. Import pandas, NumPy, Matplotlib and Seaborn; load `insurance.csv`.
2. Check structure, statistics, nulls, duplicates and column names.
3. Remove the duplicate and explore all predictors and charges through plots.
4. Create a separate encoded copy for a correlation heatmap.
5. Build model inputs with one-hot encoding and hold out 20% for testing.
6. Fit three regressors and compare prediction error and R^2 .
7. Inspect Random Forest feature importance and select the strongest recorded model.
8. Rebuild preprocessing and Random Forest as a Pipeline; save with `joblib`.
9. Reload the saved pipeline and predict a sample customer's charges.

Two different encoding steps

The integer-coded `df_encoded` is used for the correlation heatmap only. The model input `x` is created from the cleaned dataset using `pd.get_dummies(drop_first=True)`. The final saved pipeline performs categorical encoding itself.

EDA: charges & age

Original saved notebook figures; charts use the cleaned dataset.

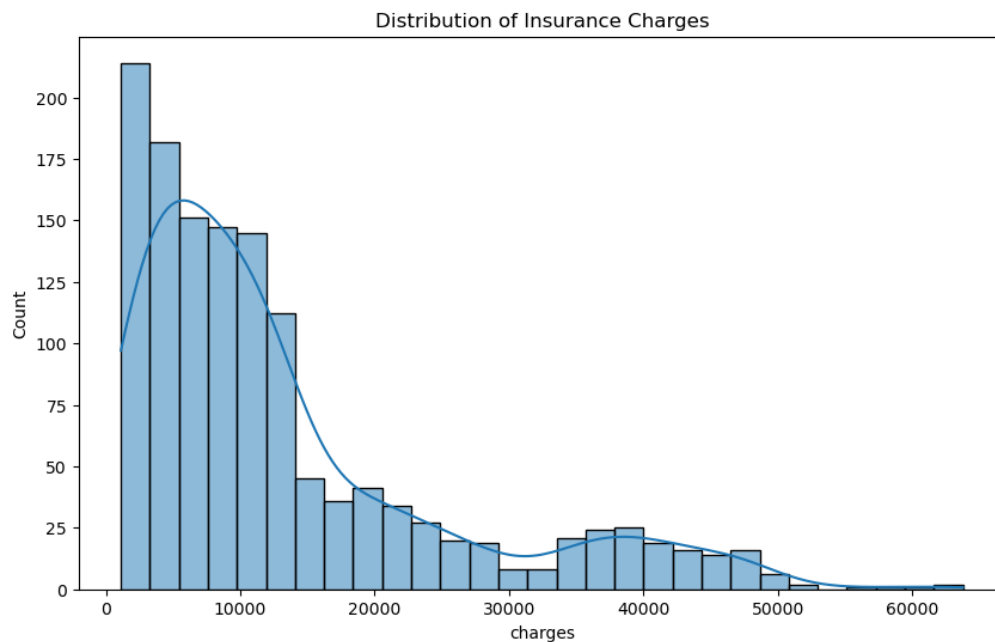


Figure 1. Distribution of insurance charges (notebook cell 12).

Charges are right-skewed: many observations have lower charges, while fewer records form a high-cost tail. The model must therefore handle a broad numerical range; a few large errors can substantially increase RMSE.

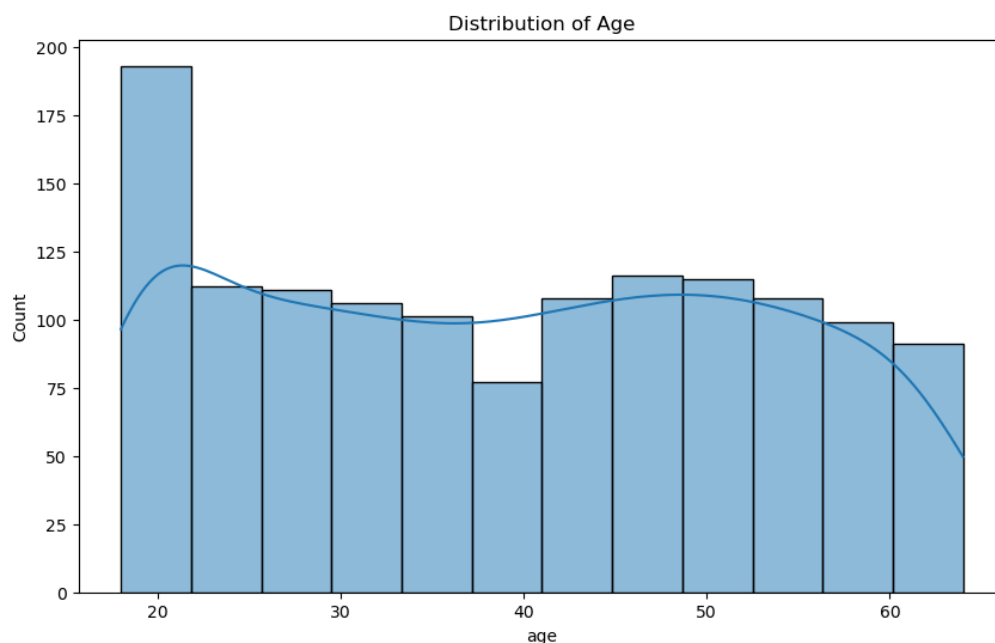


Figure 2. Distribution of age (cell 13).

Ages span 18 to 64, with a noticeable concentration at the youngest end. The dataset does not establish predictive performance for people outside this observed age range.

EDA: BMI & children

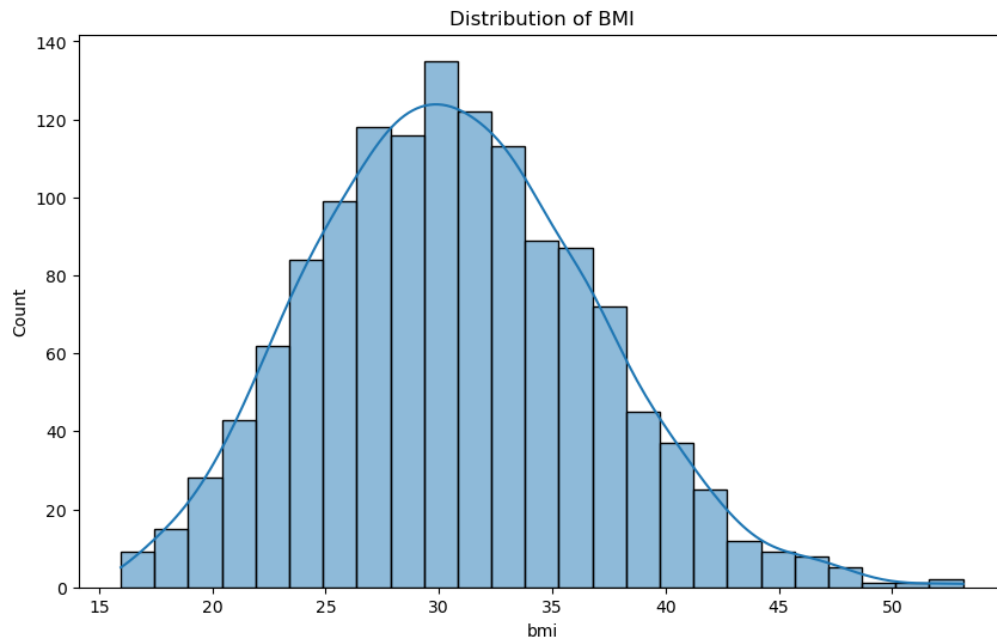


Figure 3. Distribution of BMI (cell 14).

BMI values are concentrated around the middle of the observed range, with fewer records at the extremes. The original-data mean BMI is 30.66 and median BMI is 30.40. The notebook retains these values without scaling or outlier removal.

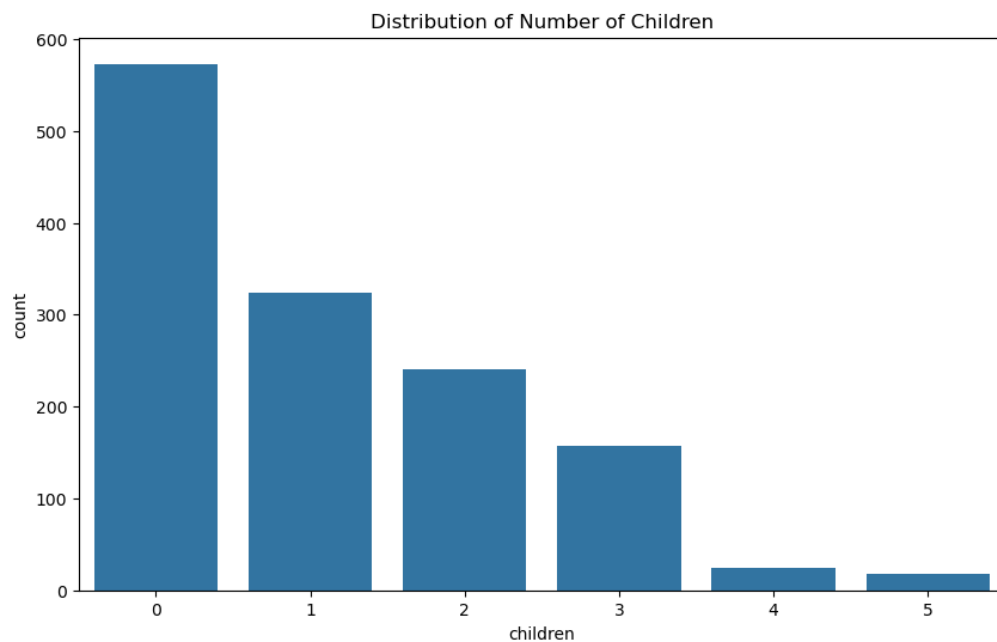


Figure 4. Distribution of number of children (cell 15).

Records with no children are the largest group. After duplicate removal, counts for 0-5 children are 573, 324, 240, 157, 25, and 18, respectively. The higher-child-count categories have relatively little data.

EDA: sex & region

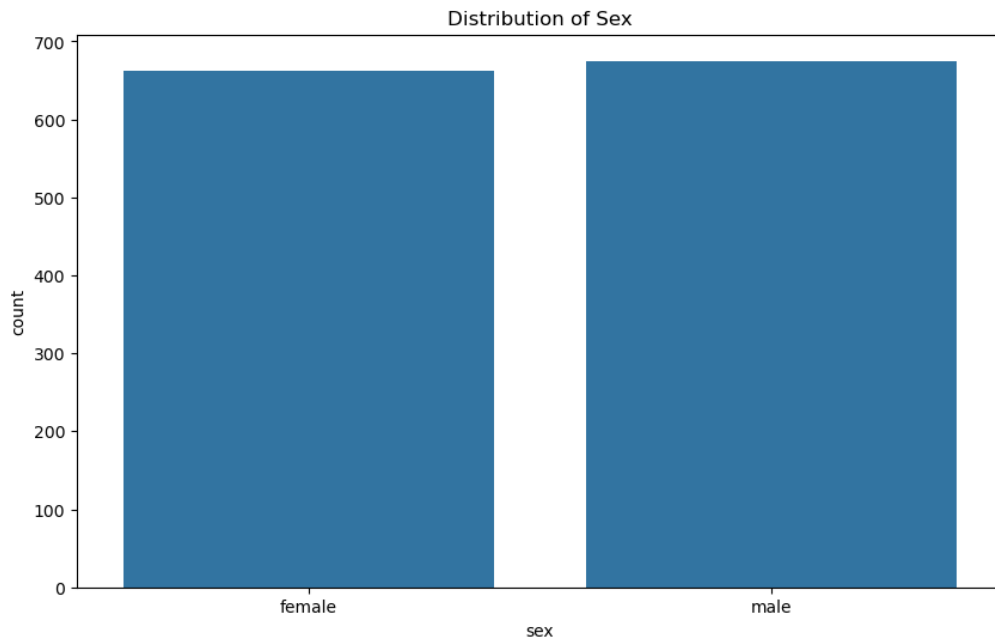


Figure 5. Distribution of the recorded sex categories (cell 16).

The cleaned dataset contains 662 female and 675 male records, giving a nearly balanced distribution for this variable. The modeling later uses female as the omitted one-hot reference category.

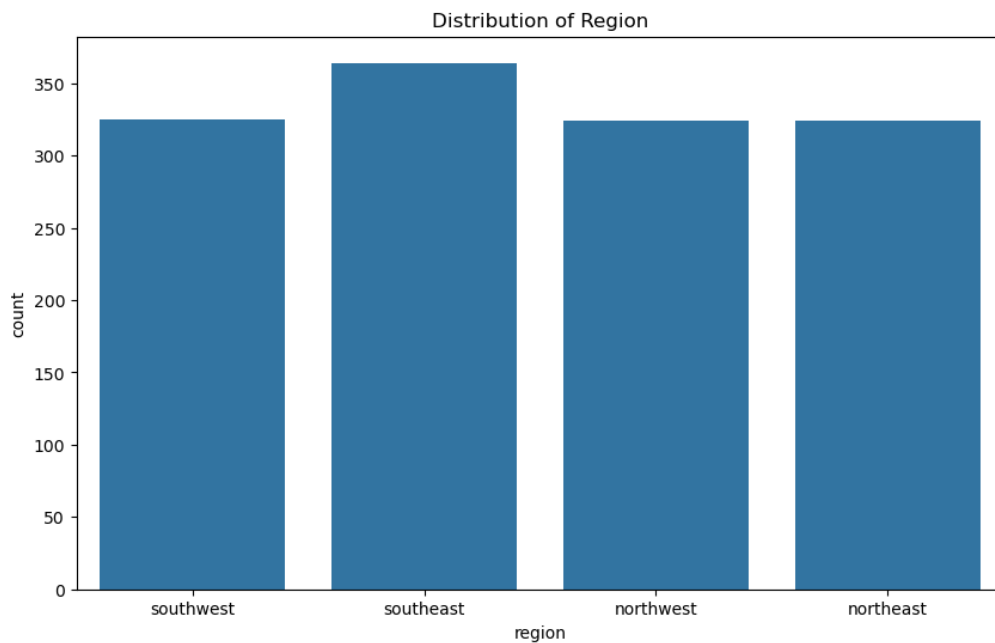


Figure 6. Distribution of region (cell 17).

Cleaned counts are southeast: 364, southwest: 325, northeast: 324, and northwest: 324. All four regions are represented; southeast has the largest count. No geographical source or population sampling method is recorded in the notebook.

EDA: smoking status & charges

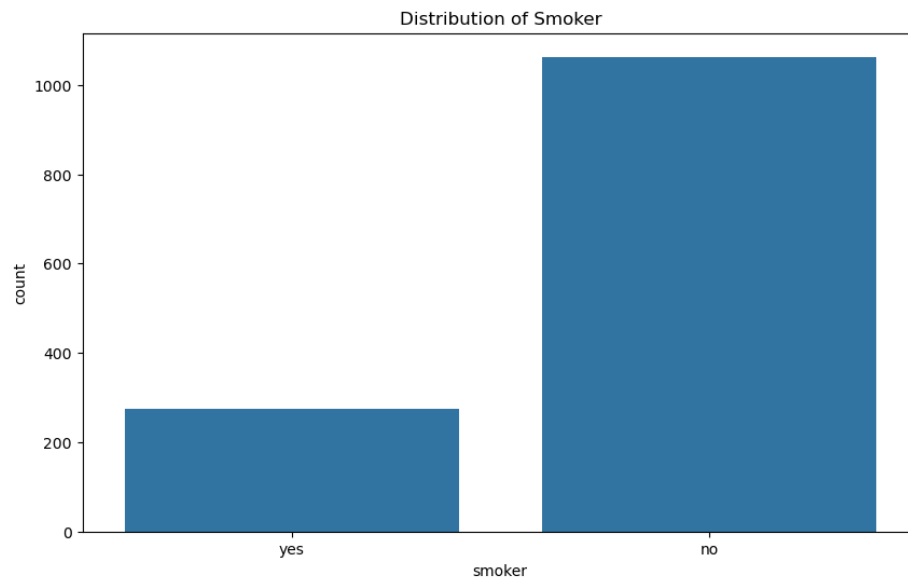


Figure 7. Distribution of smoking status (cell 18).

There are 1,063 non-smoker records and 274 smoker records after cleaning: approximately 79.51% and 20.49%, respectively. A random train/test split was used; smoking groups were not explicitly stratified.

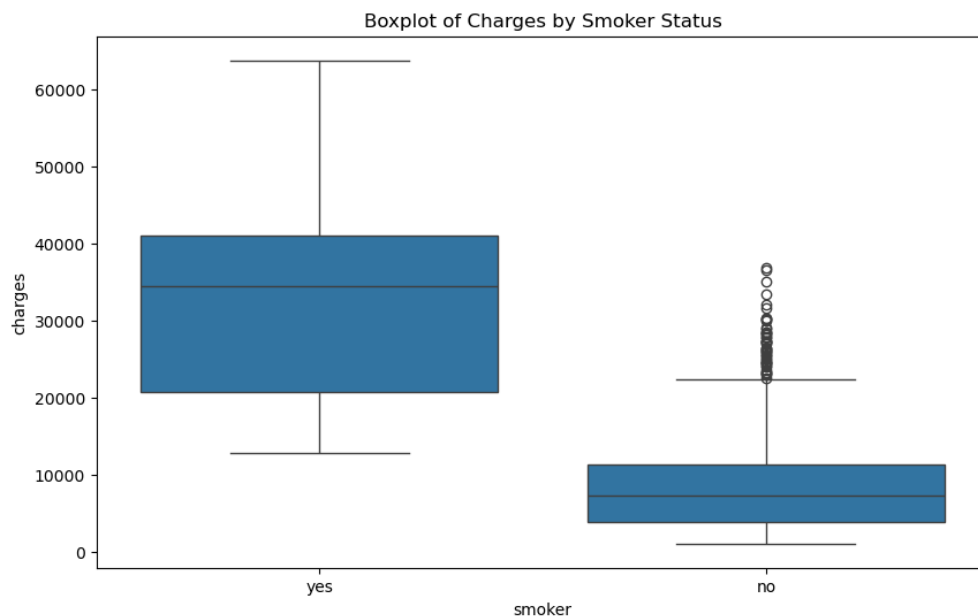


Figure 8. Charges grouped by smoking status (cell 19).

The smoker group has a visibly higher charge distribution. Supplementary calculation from the cleaned CSV gives mean charges of 32,050.23 for smokers and 8,440.66 for non-smokers, approximately a 3.80-fold difference.

This is an association in the observed dataset. The group difference and feature importance do not establish a causal effect or a price adjustment for any individual.

EDA: age & BMI relationships

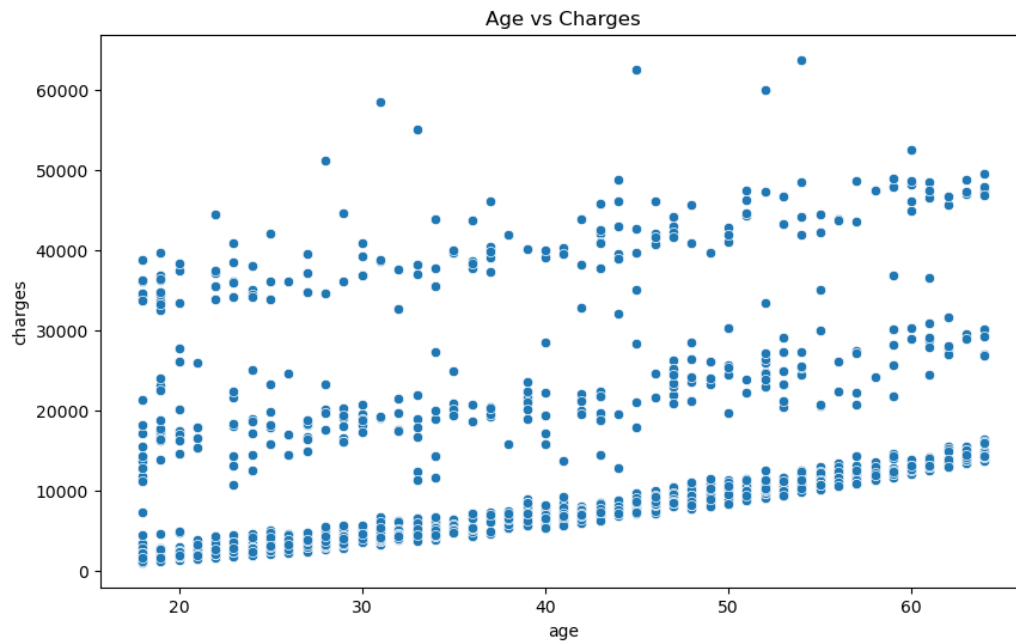


Figure 9. Age versus charges (cell 22).

Charges tend to rise with age within visible bands, while records at similar ages still vary widely. Age alone cannot explain all charge variation, making the combination of predictors important.

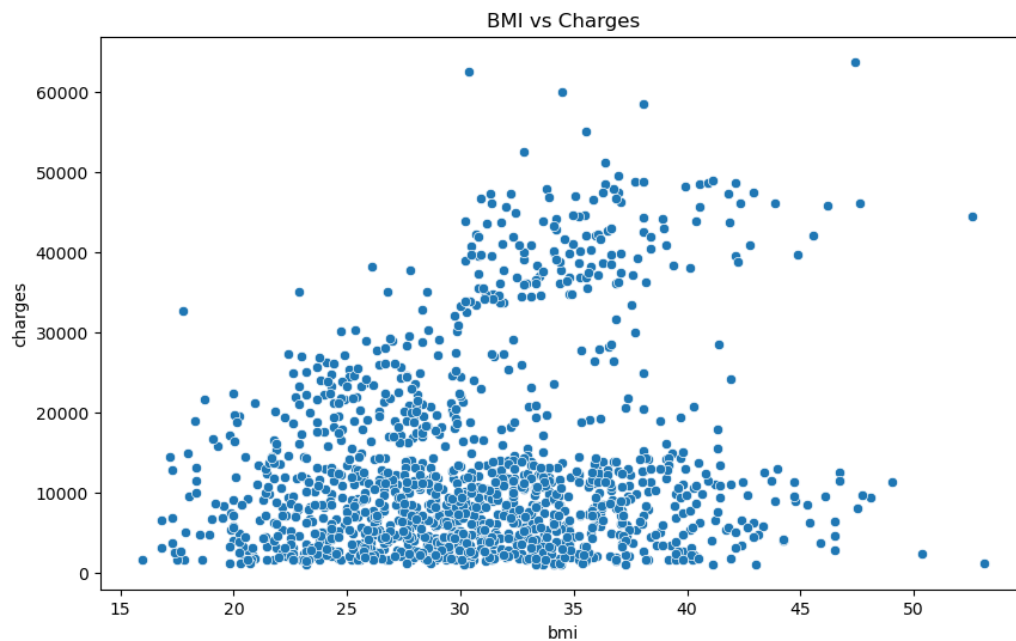


Figure 10. BMI versus charges (cell 24).

The scatter shows multiple charge levels at similar BMI values, with high-charge observations visible toward higher BMI. This structure is consistent with possible interactions that tree-based models can capture. The notebook does not plot this relationship separately by smoking status.

EDA: charges by sex & region

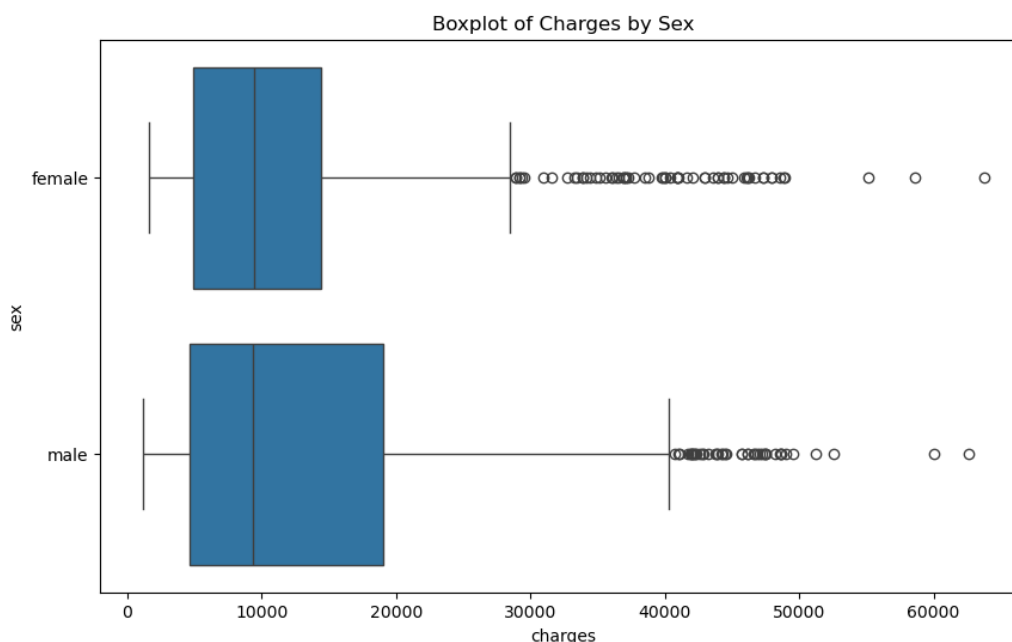


Figure 11. Charges grouped by recorded sex (cell 20).

The charge distributions for the two recorded sex categories overlap substantially. High-charge points occur in both groups. In the selected Random Forest, `sex_male` receives very low importance, which is consistent with a limited role in that fitted model.

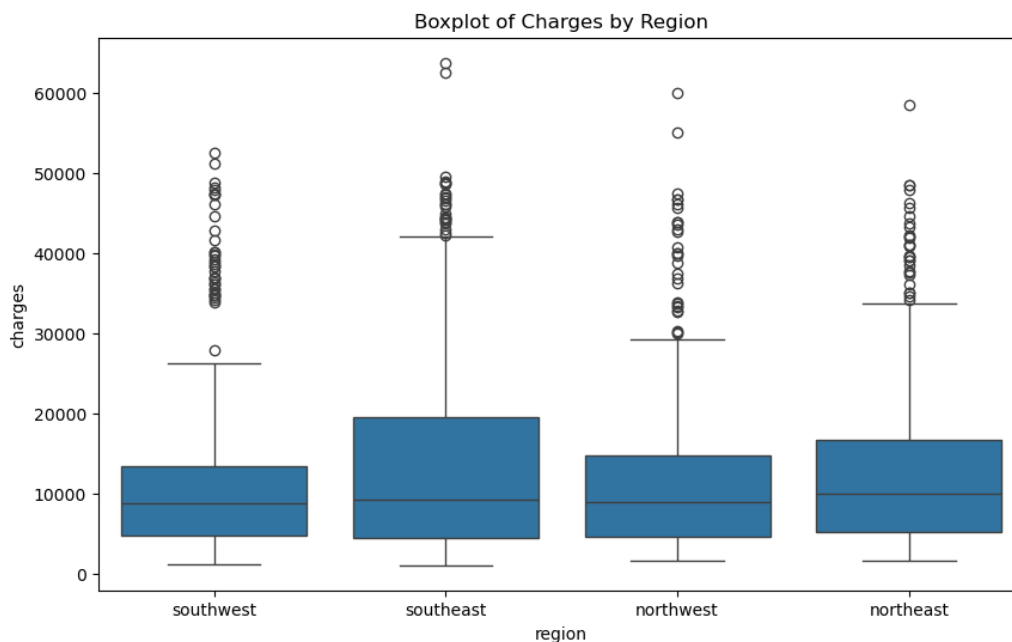


Figure 12. Charges grouped by region (cell 23).

Regional charge distributions also overlap, and southeast shows a broader high-charge spread. Such group summaries are descriptive and do not adjust for age, BMI, or smoking status; they cannot isolate the contribution of region.

EDA: children & correlations

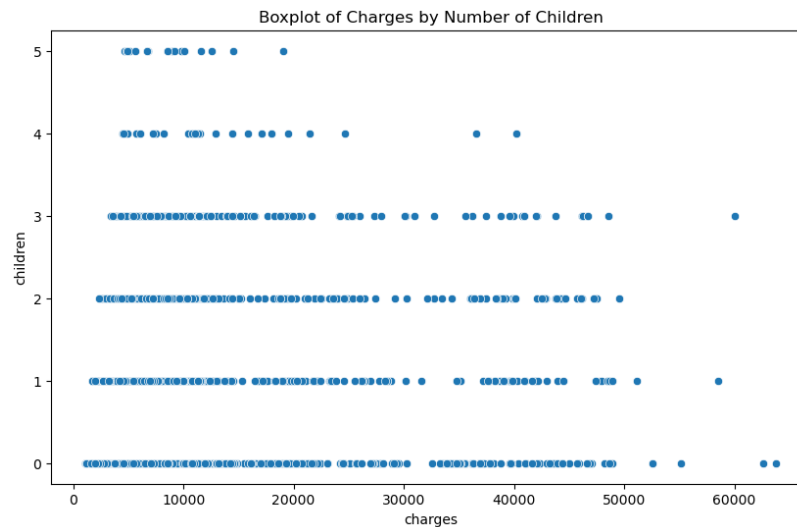


Figure 13. Scatter plot of charges and number of children (cell 21).

Charges vary widely within each child-count category. The original notebook title calls this a boxplot, but the code uses `sns.scatterplot()`; it is correctly described here as a scatter plot.

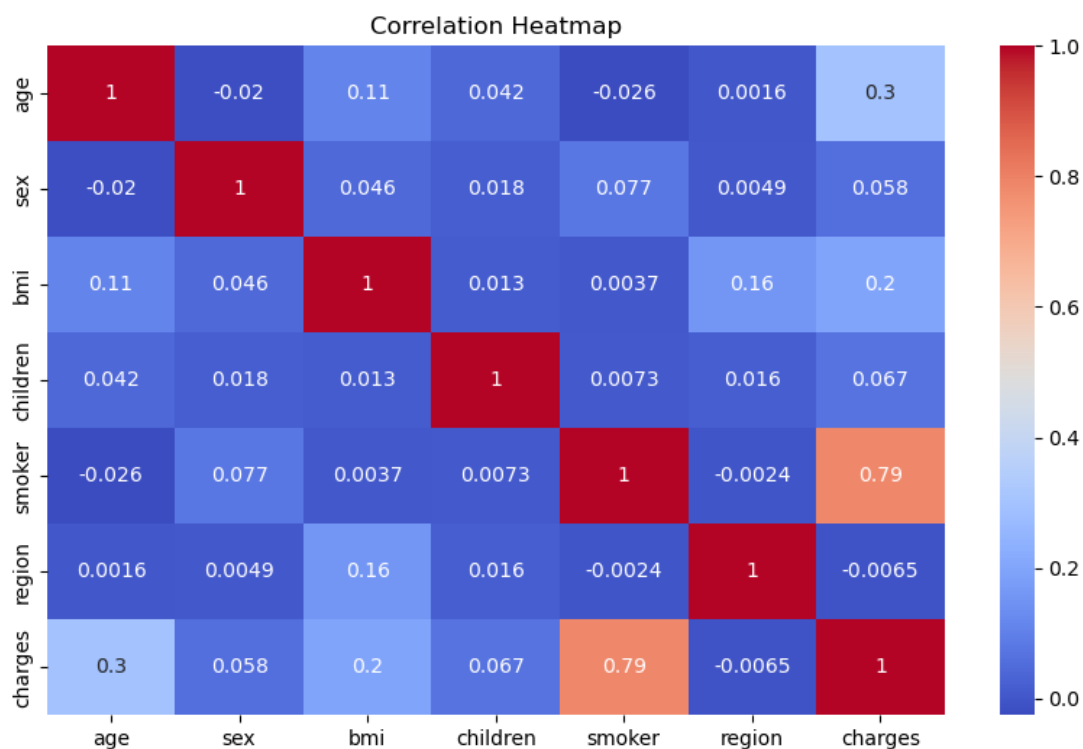


Figure 14. Correlation heatmap of the encoded dataset (cell 27).

The heatmap shows the strongest positive target correlation for smoking status (approximately 0.79), followed by age (approximately 0.30) and BMI (approximately 0.20). Linear correlation can miss nonlinear relationships and interactions.

For this exploratory heatmap, sex is mapped to female=0/male=1, smoker to no=0/yes=1, and region to category codes. Region is nominal, so the correlation with its arbitrary numeric codes should not be interpreted as an ordered geographical effect.

Feature preparation & data split

Separate predictors and target

```
x = dataset.drop('charges', axis=1)
y = dataset['charges']
x = pd.get_dummies(x, drop_first=True)
```

The target charges is excluded from the predictors. One-hot encoding converts each categorical value into binary indicator features. Dropping the first category removes redundant indicators and supplies a reference category.

MODEL FEATURE	MEANING
age, bmi, children	Three numerical variables, retained directly.
sex_male	1 for male; female is the reference.
smoker_yes	1 for yes; no is the reference.
region_northwest	1 for northwest.
region_southeast	1 for southeast.
region_southwest	1 for southwest; northeast is the reference.

Training and test split

```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

PARTITION	ROWS	PREDICTOR COLUMNS	PURPOSE
Training	1,069	8	Fit each model.
Testing	268	8	Evaluate held-out predictions.

The notebook uses the same split for the three models, which makes their recorded scores directly comparable. The fixed random seed supports repeatability of the split. No validation set or cross-validation is recorded.

Preparation detail

The earlier manual one-hot step is performed before splitting. It does not use the charges target, but the later pipeline is a cleaner pattern: fit the categorical transformer on the training data together with the model.

Regression models compared

1. Linear Regression - interpretable baseline

```
lr = LinearRegression()
lr.fit(x_train, y_train)
y_pred = lr.predict(x_test)
```

Linear Regression predicts charges as a weighted sum of the eight input features plus an intercept. It provides a useful baseline, but this specification contains no nonlinear age/BMI terms or explicit interactions. The notebook does not display fitted coefficients or inspect its residuals in detail.

2. Decision Tree Regressor - nonlinear splits

```
dt = DecisionTreeRegressor(random_state=42)
dt.fit(x_train, y_train)
y_pred_dt = dt.predict(x_test)
```

The tree divides feature space using successive decision rules and predicts from the target values within each leaf. It can capture nonlinear patterns and interactions. The constructor does not set `max_depth` or other complexity limits, and the notebook does not report training scores or tuning results.

3. Random Forest Regressor - ensemble of trees

```
rf = RandomForestRegressor(max_depth=4, random_state=42)
rf.fit(x_train, y_train)
y_pred_rf = rf.predict(x_test)
```

A Random Forest combines predictions from multiple randomized decision trees. Averaging across trees can reduce variation compared with a single tree. Here, maximum tree depth is limited to four, which constrains the complexity of individual trees.

MODEL	EXPLICIT SETTINGS IN NOTEBOOK
Linear Regression	Default constructor.
Decision Tree	<code>random_state=42</code> .
Random Forest	<code>max_depth=4, random_state=42</code> .

Other parameters are left at the installed library defaults. No grid search, randomized search, or additional model family is implemented. The results identify the best of these three tested configurations, rather than the best possible algorithm.

Evaluation & model comparison

How performance was measured

METRIC	INTERPRETATION
MAE	Mean absolute prediction error, in the same units as charges. Lower is better.
MSE	Mean squared prediction error, in squared charge units. Strongly penalizes large errors.
RMSE	Square root of MSE, in charge units. Lower is better; sensitive to large errors.
R ²	1 - residual sum of squares / total sum of squares. Measures fit relative to predicting the test-set mean. Higher is better.

Exact notebook results, rounded for readability

MODEL	MAE	MSE	RMSE	R ²
Linear Regression	4,177.05	35,478,020.68	5,956.34	0.8069
Decision Tree	2,730.63	33,281,524.62	5,769.01	0.8189
Random Forest	2,478.18	18,153,245.21	4,260.66	0.9012

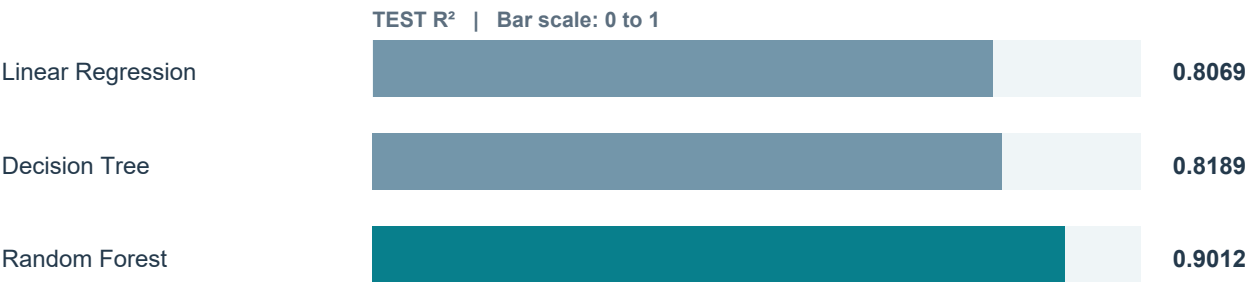


Figure 15. Comparison chart created for this report from the saved notebook scores.

Selection and practical interpretation

Random Forest has the highest R² and lowest MAE, MSE, and RMSE among the three models. Its R² of 0.9012 means it explains about 90.12% of the variation in this test set relative to the test-set mean baseline; it does not mean 90.12% of individual predictions are correct.

Compared with Linear Regression, Random Forest reduces MAE by 40.67%, RMSE by 28.47%, and MSE by 48.83%. Its average absolute error is still 2,478.18 charge units. RMSE exceeds MAE, indicating that some errors are considerably larger than typical absolute errors.

Prediction diagnostics: linear model

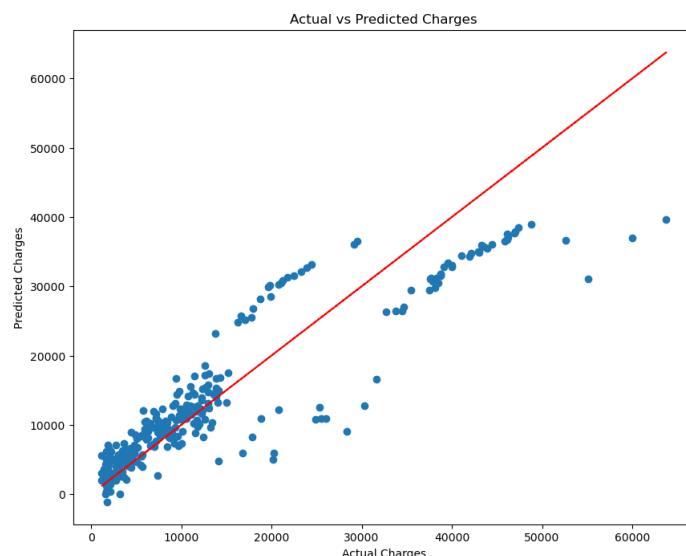


Figure 16. Actual versus predicted charges for Linear Regression (cell 44).

The red dashed line represents perfect predictions: predicted charge equals actual charge. Points above it are overestimates; points below it are underestimates. The visible spread shows where the linear baseline misses the observed charges, including several high-charge cases.

Examples from the notebook's first test predictions

ROW INDEX	ACTUAL	PREDICTED	PREDICTED - ACTUAL
900	8,688.86	8,143.69	-545.16
1064	5,708.87	5,737.12	+28.25
1256	11,436.74	14,369.31	+2,932.58
298	38,746.36	31,745.51	-7,000.84
237	4,463.21	8,962.39	+4,499.18

Rows and predictions are copied from comparison_df; residual differences are calculated for this report using the saved values. These examples illustrate individual errors rather than summarize all 268 test predictions.

Diagnostic scope

This plot and table describe Linear Regression. The notebook does not show an actual-versus-predicted plot, residual distribution, or individual test comparison table for the selected Random Forest. Those would improve the final model assessment.

Random Forest feature importance

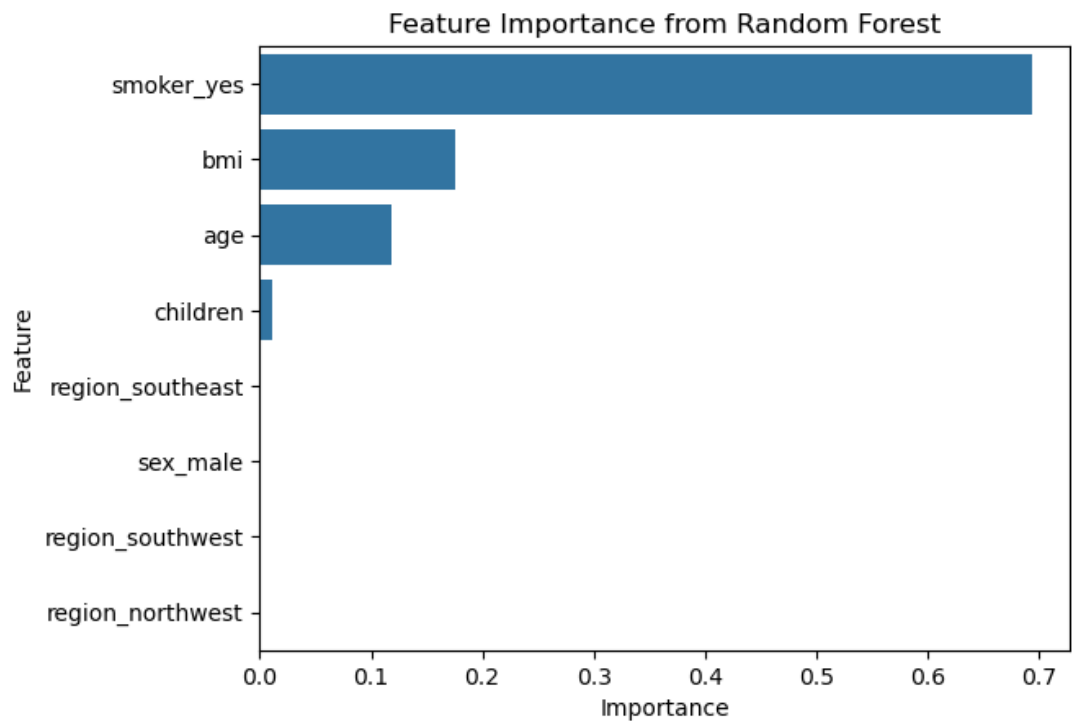


Figure 17. Random Forest feature importance from the fitted model (cell 57).

FEATURE	IMPORTANCE	APPROX. SHARE
smoker_yes	0.693633	69.36%
bmi	0.175816	17.58%
age	0.118560	11.86%
children	0.011461	1.15%
region_southeast	0.000227	0.0227%
sex_male	0.000184	0.0184%
region_southwest	0.000086	0.0086%
region_northwest	0.000032	0.0032%

Smoking status dominates this fitted model's importance ranking, followed by BMI and age. Together, these three account for approximately 98.80% of the reported importance. Children contribute much less; sex and the regional indicators contribute very little in this configuration.

These are impurity-based tree importances, normalized across features. They describe the model's use of inputs, not a percentage change in charges and not a causal effect. Values shown in the notebook are rounded, so their displayed sum can differ slightly from one.

Final prediction pipeline

Why the notebook rebuilt the selected model

The earlier models require manually encoded columns in the correct order. The final pipeline accepts the original named columns and applies the same fitted preprocessing whenever a prediction is made. This joins data preparation and the model in one reusable object.

```
categorical_columns = ["sex", "smoker", "region"]

preprocessor = ColumnTransformer(
    transformers=[
        ("categorical",
         OneHotEncoder(drop="first", handle_unknown="ignore"),
         categorical_columns)
    ],
    remainder="passthrough",
)

insurance_pipeline = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("model", RandomForestRegressor(
        max_depth=4, random_state=42)),
])
```

Training, evaluation and persistence

The raw cleaned predictors are split again with `test_size=0.2` and `random_state=42`. The complete pipeline is fitted on `X_train` and evaluated on `X_test`. Its recorded R^2 is 0.9012100890157556, matching the earlier Random Forest result.

```
insurance_pipeline.fit(X_train, y_train)
test_predictions = insurance_pipeline.predict(X_test)
joblib.dump(insurance_pipeline, "insurance_pipeline.pkl")
```

COMPONENT	ROLE
ColumnTransformer	Apply encoding only to sex, smoker and region.
OneHotEncoder	Create binary category indicators; drop the first category.
remainder="passthrough"	Retain age, BMI and children as numerical predictors.
Random Forest	Estimate charges from transformed inputs.
joblib persistence	Save and reload the fitted preprocessing and model.

`handle_unknown="ignore"` allows unseen categories to pass without an encoding failure; their indicators become all zero, which can resemble the omitted reference category. Input validation remains useful. The saved pipeline is fitted on the training portion; the notebook does not refit it on all cleaned rows.

Sample prediction & companion app

Sample testing performed in the notebook

CUSTOMER INPUT	VALUE
age	30
sex	male
bmi	25.5
children	2
smoker	no
region	southeast

```
loaded_pipeline = joblib.load("insurance_pipeline.pkl")
predicted_cost = loaded_pipeline.predict(sample_customer)[0]
```

Recorded sample estimate: 6,579.06

This value comes from the notebook's saved output after loading `insurance_pipeline.pkl`. It is an example inference, not a measured accuracy result: the sample customer has no actual charge shown for comparison.

Currency clarification

The notebook formats this output as "6,579.06 Rs". However, `insurance.csv` contains no currency field and the notebook documents no currency provenance or conversion. The number should be described in dataset charge units until the dataset's currency and target definition are confirmed.

Supplementary interface found alongside the notebook

The companion `app.py` contains a Streamlit form for age, sex, BMI, children, smoker status, and region. On submission, it builds a one-row pandas DataFrame, calls the loaded pipeline, and displays the estimated charges.

The form restricts age to 18-64 and children to 0-5, matching observed ranges. Its BMI range is 15.0-55.0, slightly wider than the dataset range of 15.96-53.13. The app displays a numeric estimate without a currency symbol.

This is a source-code description of the adjacent app, not a claim that it was run or deployed. The notebook itself completes the pipeline save/load and sample inference steps.

Limitations & future improvements

What the current evidence does not establish

- **Generalization across splits:** Results use one random 80/20 split. No cross-validation or repeated-split variability is recorded.
- **Unbiased final selection estimate:** The same test set is used to compare models and select Random Forest. A further untouched holdout would provide a stronger final assessment.
- **Hyperparameter search:** `max_depth=4` is specified, but the notebook records no tuning process or evidence that this setting is optimal.
- **Final-model error patterns:** No Random Forest residual plots, subgroup metrics, prediction intervals, or training/test gap are shown.
- **Dataset provenance:** Collection method, target period, verified currency, and population representativeness are not documented.
- **Production behavior:** A serialized pipeline and app source are present, but hosted deployment, operational monitoring, and validation on new data are not demonstrated.

Suggested next steps - not implemented in this notebook

- Use cross-validation on the training data to compare and tune models, then evaluate once on an untouched holdout.
- Add a mean-prediction baseline and evaluate Random Forest residuals and actual-versus-predicted plots.
- Report MAE and RMSE by smoking group, age band, and region to expose uneven performance.
- Compare alternative tree ensemble configurations and assess age/BMI/smoking interactions through controlled experiments.
- Use held-out permutation importance or another validated explanation method to check the importance ranking.
- Document dataset origin, currency and the definition of charges; validate category inputs and observed numerical ranges.
- If deployed, version the pipeline and dependencies, test the interface, and monitor data drift and error on new labeled records.

How to present the result

A defensible claim is: "Random Forest gave the strongest recorded held-out performance among the three tested models on this dataset." Avoid converting R^2 into an individual prediction accuracy guarantee.

Conclusion & reproducibility notes

Final project conclusion

The project completed an end-to-end regression workflow for estimating medical insurance charges. It inspected and cleaned the dataset, explored distributions and feature relationships, encoded categorical predictors, compared three regressors, and packaged the selected approach into a saved prediction pipeline.

Random Forest with `max_depth=4` performed best on the recorded 268-row test set: $R^2 = 0.9012100890$, MAE = 2,478.1816, and RMSE = 4,260.6625. Smoking status was the largest contributor to its feature importance, followed by BMI and age. A reloaded pipeline produced a sample estimate of 6,579.06.

Environment reported in the notebook

COMPONENT	RECORDED VERSION / USE
Python	3.13.5, Anaconda distribution
scikit-learn	1.6.1
pandas	2.2.3
NumPy	2.1.3
joblib	1.4.2
Matplotlib / Seaborn	Used for visualization; versions not printed.
Streamlit	Used in companion app; version not pinned.

Files and their roles

FILE	ROLE
Insurance_Cost_Prediction.ipynb	Primary source: code, saved outputs and plots.
insurance.csv	Local dataset; used to verify row and category counts.
insurance_pipeline.pkl	Fitted preprocessing and prediction object, saved in the notebook.
app.py	Supplementary Streamlit inference interface, inspected as source.
requirements.txt	Adjacent dependency pins; Streamlit is unpinned.

The notebook prints its environment versions at cell 62. Its working directory must contain `insurance.csv` to rerun the recorded load operation. The sample test expects `insurance_pipeline.pkl` in the working directory. The application expects the same saved pipeline.

Report evidence and calculations

All model metrics, feature importances, versions and sample predictions are taken from saved notebook outputs. Sixteen original figures are included, plus one report-created R^2 comparison chart. Additional CSV counts, group means, relative error reductions and example residuals are labeled as report calculations. No new model training, inference, deployment or externally sourced dataset claim was performed for this document.